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Mini project

Malicious URL Detection: An AI-Based Approach

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Module: Data security

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Abstract

This report presents the development and evaluation of an AI-based model for detecting malicious URLs. Malicious URLs pose significant cybersecurity threats as they are commonly used in phishing attacks and malware distribution. We implemented and compared multiple machine learning and deep learning approaches for classifying URLs as benign or malicious, including Logistic Regression, Random Forest, XGBoost, and Long Short-Term Memory (LSTM) neural networks. Using the Malicious URLs dataset from Kaggle containing 651,191 URLs, we achieved a maximum accuracy of 97.82% using a Random Forest classifier. Our findings demonstrate the effectiveness of machine learning techniques in identifying malicious URLs and provide insights into the trade-offs between model accuracy and computational efficiency.

1 Introduction

1.1 Context and Problem Statement

In today's digital landscape, malicious URLs represent a significant cybersecurity threat. These URLs are commonly employed in phishing attacks designed to steal sensitive information or distribute malware. The ability to detect these malicious URLs accurately and efficiently is crucial for protecting users and systems from various cyber threats.

Traditional approaches to URL detection often rely on blacklisting or rule-based systems, which struggle to keep pace with the rapidly evolving tactics of cyber attackers. Machine learning and deep learning techniques offer promising alternatives, as they can identify patterns and features that might not be apparent to human analysts or rule-based systems.

1.2 Project Objectives

The main objectives of this project are:

1. Develop an AI-based model for detecting malicious URLs
2. Compare different machine learning and deep learning approaches
3. Identify the most effective techniques for this classification task
4. Analyze the performance-efficiency trade-offs among different models

1.3 Dataset Description

For this project, we used the "Malicious URLs dataset" from Kaggle, which contains 651,191 URLs classified into four categories:

- Benign: Normal, legitimate URLs (65.74%)
- Phishing: URLs designed to steal sensitive information (14.45%)
- Defacement: URLs pointing to compromised websites (14.81%)
- Malware: URLs that distribute malicious software (4.99%)

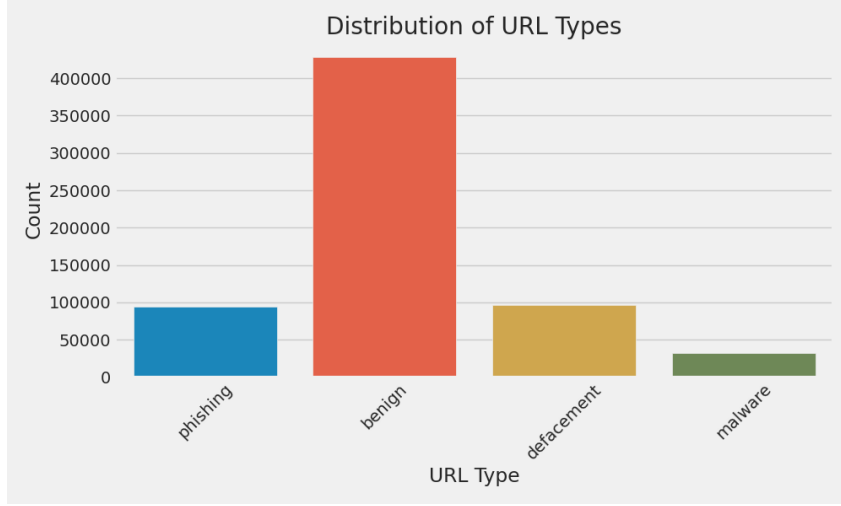


Figure 1: Distribution of URL Types in the Dataset

2 Methodology

2.1 Data Exploration and Preprocessing

We began by exploring the dataset to understand its characteristics and distribution. Key findings include:

- The dataset contains 651,191 URLs with no missing values
- Class distribution is imbalanced, with benign URLs constituting 65.74% of the data
- URL lengths vary significantly across different categories

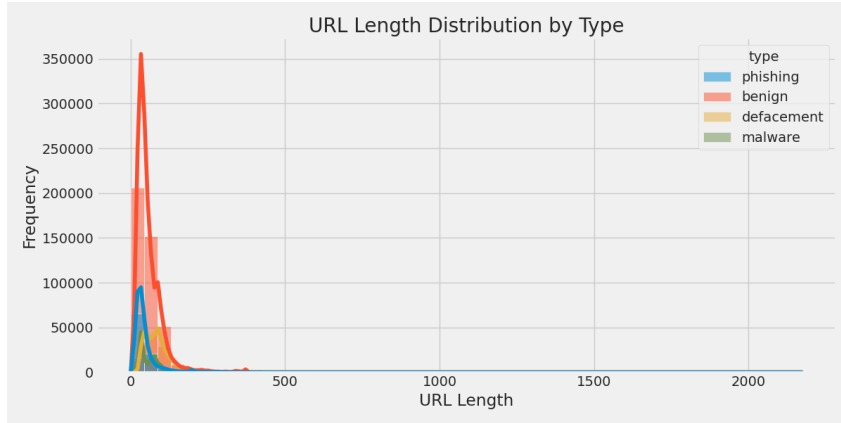


Figure 2: Distribution of URL Lengths by Class

2.2 Feature Extraction

We utilized several techniques for feature extraction:

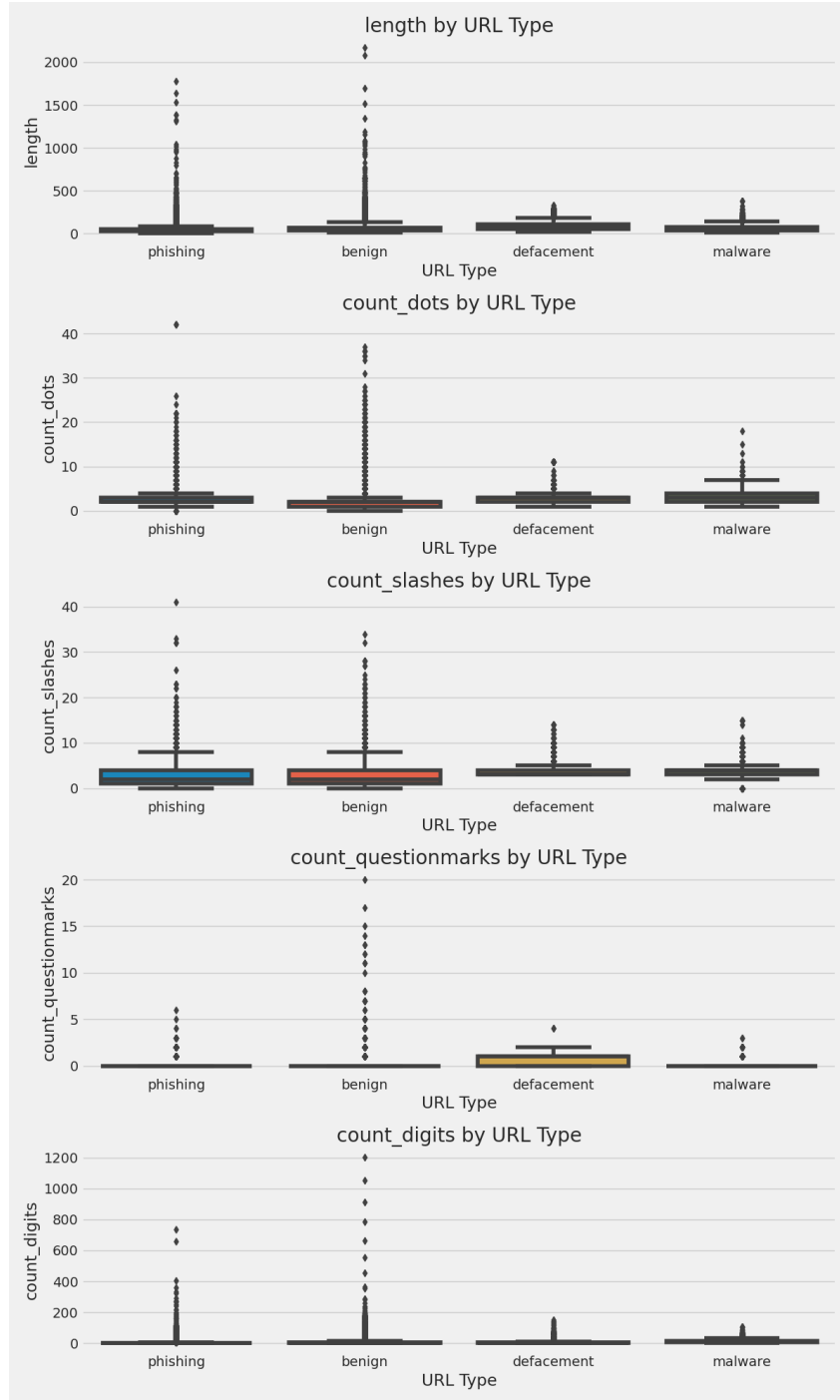


Figure 3: URL Feature Characteristics by Type

1. **TF-IDF Vectorization:** We employed character-level n-grams (3-5) to capture patterns in the URL structure. This method transforms the text of the URLs into numerical feature vectors, considering character sequences rather than whole words.
2. **URL-Specific Features:** We extracted domain-specific features including:
 - URL length
 - Count of special characters (dots, hyphens, slashes, etc.)
 - Presence of IP addresses

- Count of digits and special symbols

2.3 Model Implementation

We implemented and compared four different models:

1. **Logistic Regression**
2. **Random Forest**
3. **XGBoost**
4. **LSTM (Long Short-Term Memory)**

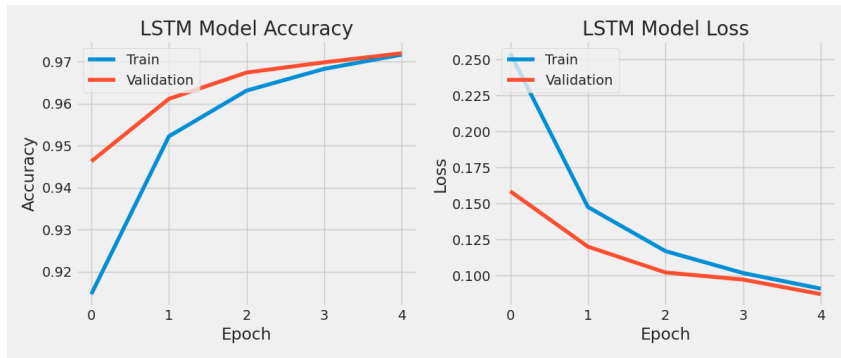


Figure 4: Training History of LSTM Model

3 Results and Discussion

3.1 Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score	Training Time (s)
Logistic Regression	96.48%	96.43%	96.48%	96.43%	186.91
Random Forest	97.82%	97.80%	97.82%	97.80%	3041.13
XGBoost	96.53%	96.48%	96.53%	96.46%	561.42
LSTM	97.33%	97.31%	97.33%	97.30%	~260

3.2 Confusion Matrices

3.3 Feature Importance Analysis

3.4 Performance-Efficiency Trade-offs

Our experiments reveal a clear spectrum of computational cost versus classification accuracy. The Random Forest model achieved the highest accuracy (97.82%), but required the longest training time (over 3000 s), making it less suitable for scenarios where rapid model retraining or resource constraints are critical. In contrast, the LSTM network

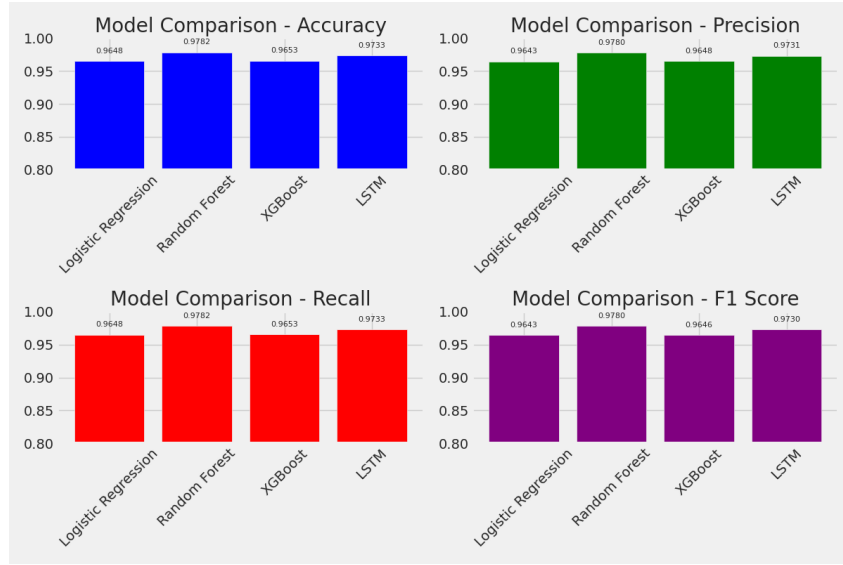
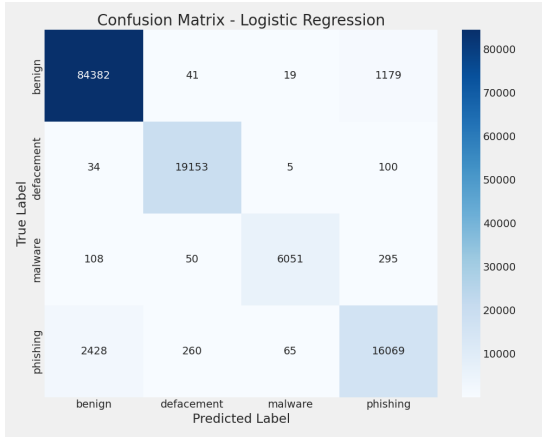
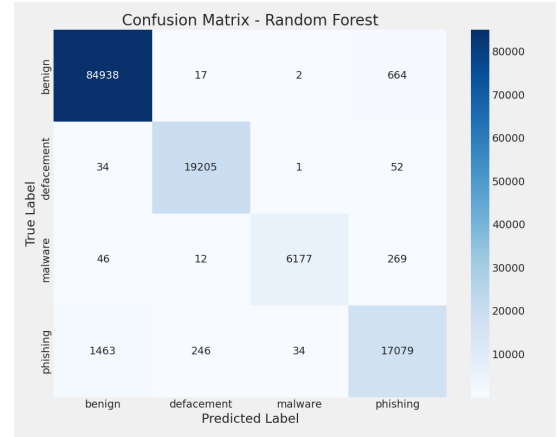


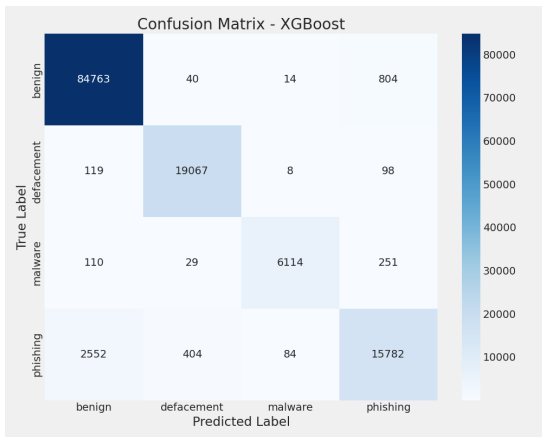
Figure 5: Comparison of Model Performance



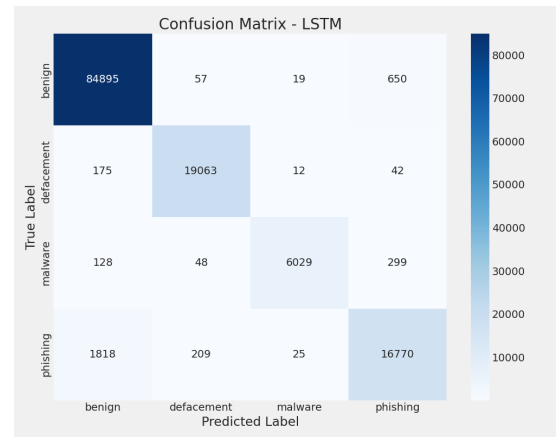
(a) Logistic Regression



(b) Random Forest



(c) XGBoost



(d) LSTM

Figure 6: Confusion Matrices for All Models

delivered near-top accuracy (97.33%) with only 260 s of training, benefiting from GPU acceleration and sequence modelling of URL character patterns. Logistic Regression and

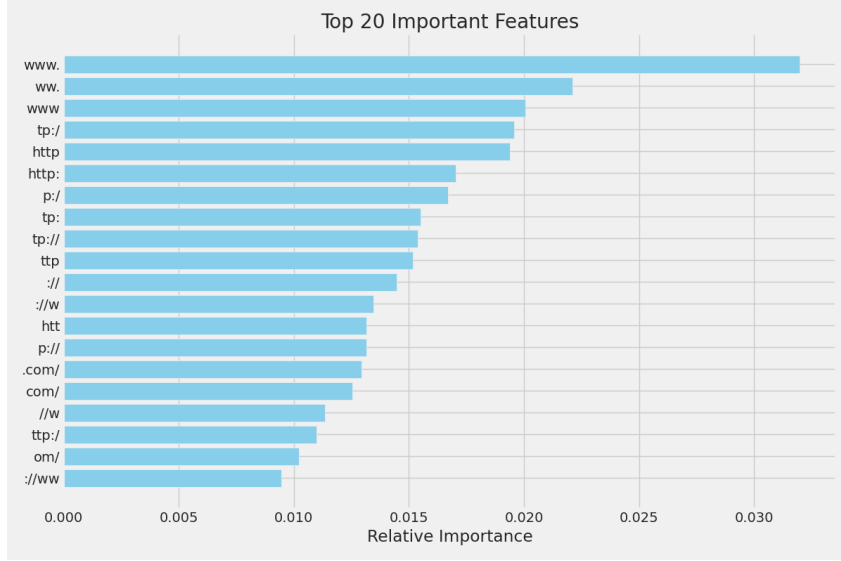


Figure 7: Feature Importance from Random Forest

XGBoost both trained in under 600 s; Logistic Regression (96.48%) is appealing for its simplicity and interpretability, while XGBoost (96.53%) provides marginally better performance at the cost of slightly longer fit times. Thus, when deploying in a production environment, the choice of model should balance the required level of detection accuracy against available compute resources and latency requirements.

3.5 Discussion

The high accuracy across all four methods demonstrates that both traditional ML and deep learning approaches can effectively distinguish malicious from benign URLs given rich feature representations. Character-level TF-IDF proved powerful, capturing subtle token patterns that often indicate phishing or obfuscation tactics (e.g. excessive hyphens, IP-literal domains). Domain-specific features (URL length, special-character counts, presence of digits) further boosted performance, especially for tree-based models.

However, certain limitations remain. Our dataset, while large, may not fully reflect zero-day malicious URLs that exploit novel obfuscation or multimedia-based payload delivery. The imbalance in class distribution also suggests that minority classes (e.g. defacement and malware) could benefit from targeted oversampling or cost-sensitive learning to avoid bias. Finally, although LSTM captured sequential dependencies, it required careful tokenization and more hyperparameter tuning to approach the Random Forest’s accuracy.

4 Conclusion and Future Work

In this mini-project, we developed and benchmarked four AI-based methods for malicious URL detection. A Random Forest classifier achieved the best overall accuracy (97.82%), while an LSTM model offered a compelling trade-off between speed and performance. Logistic Regression and XGBoost delivered strong baseline results with minimal complexity.

For future work, we propose:

- **Online learning and adaptability.** Integrate data streaming to incrementally update models as new URLs are observed, improving resilience to novel attack vectors.
- **Ensemble and hybrid architectures.** Combine interpretable tree-based models with sequence-aware deep nets in a stacked ensemble to further boost detection rates.
- **Feature augmentation.** Incorporate additional metadata such as WHOIS registration age, hosting IP reputation, and DNS-level features to capture attacker infrastructure signals.
- **Adversarial robustness.** Evaluate model performance under adversarial URL obfuscation (e.g. homoglyphs, zero-width chars) and apply defenses like adversarial training or input sanitization.

Overall, our results validate that machine learning and deep learning can form the core of an effective malicious URL detection system, and further enhancements will strengthen defence against evolving web threats.